

Joseph Voss

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EDUCATION

Bachelor of Science, Mechanical Engineering
University of Texas at Austin

Aug 2014 – May 2018

EXPERIENCE

Systems Reliability Engineer

Cloudflare

Aug 2021 – Present

- Rotated through operational oncall covering 310+ edge colos globally, coordinating cross-team incident response for customer-impacting events.
- Led incident response and authored post-mortems for several high-severity events. Participated in the fleet-wide CDN latency regression documented in a public Cloudflare blog post.
- Designed and built an automated colo recovery system using Temporal workflows, orchestrating the full recovery sequence for offline/degraded edge colos. Contributed workflow activities for IPMI, Redfish, and switch configuration to the broader edge automation platform.
- Designed and built an AI-powered operational memory system using sentence-transformer embeddings, a local vector database, and LLM-generated summaries to surface relevant alert context during oncall. Developed low-friction and feedback-driven institutional knowledge store for agents.
- Implemented Redfish-over-management-switch IPMI operations enabling power cycling of offline servers from chat commands, eliminating the need for manual hands-on intervention in offline colos.
- Built a just-in-time access system for edge jumphosts providing temporary, audited access grants with automatic expiration. Extended to support FedRAMP High colos via dedicated GCP jumphosts with firewall enforcement.
- Extended automated server reboots to a partnership fleet without anycast infrastructure, building custom traffic management tooling to safely drain datacenters for reboots.
- Led end-to-end migration of the Salt CI pipeline from TeamCity to GitLab CI. Reduced CI runtime, standardied dependency management, and shipped complementary tooling.
- Designed and delivered a self-service lab metal reservation portal, replacing a static document with automated lifecycle management including time-based expiration and automatic cleanup.
- Authored RFC and drove implementation of upstream Debian repository snapshotting, hardening the fleet's apt supply chain with staged, health-mediated rollout instead of direct upstream pulls.

HPC Systems Engineer

Oak Ridge National Laboratory

Jun 2018 – Aug 2021

- Provisioned Andes, new 700 node HPC cluster — largest commodity cluster procured by ORNL to date.
- Wrote custom tool (Anchor) to boot HPC machines from container images. Transitioned several large-scale systems to use it.
- Developed eBPF wrapper (greggd) to compile and load Linux kernel profiling programs and output data to Kafka.
- Created Helm charts and automated pipelines to move system services to Kubernetes.
- Developed CI pipeline to stage Puppet changes on bare metal servers.
- Reviewed proposals for new HPC systems; assisted in their provisioning and deployment.
- Extended Let's Encrypt Golang projects to create in-house certificate issuer for host authentication bootstrapping.
- Used Puppet to configure and manage large-scale systems.
- Contributed to open source projects to improve system health and monitoring; used to create load-balanced and highly available data transfer cluster.
- Helped lead a team of student interns in the '19 Student Cluster Competition.

SKILLS

Languages: Golang · Python · Rust · Bash · Typescript

Infrastructure & Configuration Management: Saltstack · Puppet · Terraform · Kubernetes · Nomad · cdk8s / Helm · Debian / RHEL Linux · GCP · Cloudflare Workers

Observability, Reliability & Systems: Prometheus · Grafana · PagerDuty · Consul · Vault · Temporal · Airflow · Redfish / IPMI · eBPF · Kafka

CI/CD & Developer Tooling: GitLab CI · Jenkins · Git / Jujutsu

AI-Assisted Development & ML: Opencode · Pi-coding-agent · RAG / Vector Databases · Sentence Transformers · LLM Orchestration